

Dean S. Keeble

Principal Scientist, Scientific Software Developer, Physicist, Materials Scientist.

[github](#) | [orcid](#)

Currently

Principal Scientist at Diamond Light Source

Personal Statement

I'm a scientist whose passion is enabling experiments that require high energy photons. My background is in materials science and scientific software, so I am a curious problem-solver who enjoys understanding the fundamentals of challenges. I am a strong communicator and excel at translating between technical domain-specific languages. I've managed scientists, technicians, and graduate software engineers; I've led high-profile projects; and represented the facility internationally.

Technical Skills

Python, Git(hub/lab), Jira/Confluence, Linux, C++, slurm, UGE, bash, neovim, SQL, marp; Windows & Office.

Research interests

Total scattering theory, pair distribution function, synchrotron experimental design including using liquids, gases, electric fields, extremes of pressure and temperature.

Publications

73 Publications; 2428 Citations; h-index 25 (as of Aug 2026)
Full list of publications available on [orcid](#).

Occupation

Principal Beamline Scientist, Diamond Light Source

Aug 2024 - present

[instrument management](#) / [project leadership](#) / [scientific strategy](#)

Responsible for the I15-1 beamline; line management, budget responsibility, scheduling, strategy and operational decision making

Senior Software Scientist, Diamond Light Source

Jan 2023 – Aug 2024

[project management](#) / [leadership and supervision](#) / [facility representation](#)

Leading high profile software projects; representing Diamond in collaborations with other international facilities; technique development for novel data collections and processing algorithms; management of PhD students and graduate scheme projects

Data Analysis Scientist, Diamond Light Source

May 2020 – Dec 2022

[technique development](#) / [algorithm development](#) / [process automation](#)

Developing and maintaining analysis applications and libraries to support beamline users; deployment of new automated routines

Beamline Scientist, Diamond Light Source

March 2015 – May 2020

[customer liaison and training](#) / [facility advocacy](#) / [equipment purchase and commissioning](#)

Being first point of contact for user experiments, covering safety, user training, scientific consultation and experiment optimisation; development of data collection and analysis software; purchasing and commissioning of beamline hardware

Post-Doctoral Research Associate, Diamond Light Source August 2014 – March 2015
facility establishment / instrument commissioning

Installation, testing, and commissioning of instrument components; design and development of the instrument's bespoke on-the-fly data analysis software, designing data correction algorithms and user experience

Science City Research Alliance Research Fellow, University of Warwick 2009 – July 2014

Post-Doctoral Research Assistant, University of Warwick
academic research / scientific publications and presentation

Investigating the structure-property relations in various single crystal oxides including Lithium Niobate

Education

Department of Physics, University of Warwick 2005 - 2009
Ph.D. An investigation of the relationship between the structure and properties of some stacked perovskites

Department of Physics, University of Warwick 2001 - 2005
MPhys 2:1